



CAMBRIDGE

Sample of Work

Driving Down Ireland's Transport Emissions

Progress in decarbonising Ireland's transport sector has not kept pace with the country's ambitious climate targets. The Environmental Protection Agency projects that the sector will miss its 2030 Greenhouse Gas emissions reduction target by 21% [1]. This essay overviews and broadly evaluates current policy measures in place to reduce transport sector emissions. Possible improvements are discussed, including a Cap-and-Trade emissions trading system.

Michael Kerins

December 10, 2024

Ireland's adult population widely accept that Climate Change is occurring, and agree the country has a responsibility to act [2].

The forecasted global cost of inadequate climate action has been comprehensively modelled, and consists of profound economic, health, and biodiversity impacts. In Ireland, the Economic and Social Research Institute has projected that sea level rise under a moderate warming scenario will cost the country €2 billion in 2050 [3].

Transport contributed 21% of Ireland's Greenhouse Gas (GhG) emissions in 2023 and was the second largest sectoral emitter behind Agriculture [4]. The country has committed to a 51% reduction in GhG emissions by 2030¹, and achieving the sectoral target of 50% in the transport sector will prove critical. Progress in the transport sector has lagged behind others, and policy measures must be reviewed and improved, so that Ireland can do its part in the global effort to tackle climate change.

Various policy measures have been introduced to encourage a move away from fossil fuels in the transport sector. The main measures include:

- A Carbon tax levied on fuel suppliers in Ireland, aiming to reduce consumption.
- Gradual electrification of the public transport network.
- Electric vehicle (EV) grants of up to €3500.
- Reduced public transport fares, and a fare cap integrated across modes of transport.
- Extension of the rural bus network, with 50 settlements newly connected in 2022.
- Regulation supporting bioethanol fuel blending in petrol, and biodiesel.

The measures have contributed to a 4% reduction in GhG emissions from the transport sector since 2018 [5]. However, emissions from the sector remain largely coupled with economic and social activity. They have increased by an annual average of 5% over the last three years, after a drop in 2020 due to travel restrictions during the pandemic. The carbon tax increased by €7.50 per annum over the same period, indicating it may not effectively reduce fossil fuel consumption.

A critical determinant of the carbon tax's ability to reduce transport sector emissions is the availability of alternative low-carbon modes of transport. For many people in rural Ireland this is not the case. Although progress has been made, there remains limited access to reliable and regular public transport for many rural communities. The issue is particularly evident in the country's north-west, which is not connected to the rest of the country by rail. As well as this, exceptional inflation in recent years has reduced purchasing power, negating the effect of EV grants. The 'Climate Change in Ireland's Mind' study conducted in 2021 and again in 2023 illustrates these two points [6, 2]. Public support for higher tax on petrol and diesel cars has decreased by 13 percentage points to 51% over the period. Additionally, support for this policy is at a minimum in the north-west counties of Sligo and Donegal, and generally low across rural counties.

The above factors contribute to a low elasticity of fuel demand. The result is that the carbon tax may be unduly eroding household expenditure while having little effect on emissions. A second critical issue is the equity of the tax, which is potentially disproportionately burdening Ireland's rural population, who have no alternative but to continue purchasing fossil fuels. This second issue undermines Ireland's commitment to a socially inclusive 'Just Transition' towards climate neutrality.

A policy option worth exploring is a Cap-and-Trade system for carbon emissions. This system provides a transparent method of forcing a pre-designated emissions reduction while allowing markets to determine the cheapest actions to make the reduction. The feasibility of these systems has been proven by the success of the European Emissions Trading System (ETS), which covers heavy emitting industries in Europe. Ireland is a participant of this system, and 109 installations in the country are included. The European ETS has helped reduce emissions from electricity and heat generation as well as industrial production by 37.3% of their 2005 levels [7].

¹relative to 2018 levels.

Commercial goods transport accounted for 43% of road transport emissions in 2022 and could be an ideal candidate industry for a Cap-and-Trade system in Ireland. The main barrier to implementing this kind of system is the availability of quality carbon emissions data from each participating business. This issue is being addressed by new Corporate Sustainability Reporting regulations, which require large companies in Ireland to begin reporting on their carbon emissions from 2025, with small and medium enterprises obliged to report from 2028.

Tackling commercial emissions using this system would finance further government investment in the public transport network and electric vehicle grants. Furthermore, it would enable a system of policy measures that focus specifically on domestic transport emissions.

Ireland has had success reducing domestic heating emissions, which have fallen by an average of 9% over the last three years. The combination of a carbon tax, and heavily subsidized low-carbon heating alternatives drove this success. Transport policy must mirror this and focus on providing low-carbon transport alternatives, with a significant concentration of effort in rural Ireland. Expansion of the country's rail network is worth considering and could be an excellent use of the recent €14 billion tax windfall from Apple. Expansion of the rural bus network has already been successful, with record customer numbers being recorded in rural Ireland; and this program of work must continue at pace [8]. With alternatives in place, the carbon tax has significantly more potential to encourage a reduction in emissions.

One final measure with the potential to encourage behavioural changes, are Awareness Campaigns. Ireland's recent General Election saw the country's Green Party lose all but one seat in parliament, perhaps indicating that Climate Change has slipped from the country's consciousness. Continuous effort must be made to increase the visibility of climate action to the people of Ireland. A quote from Sir David Attenborough in 2020 still rings true: "Saving our planet is now a communications challenge. We know what to do, we just need the will." The collective will of Ireland must be raised and maintained if the country is to rise to the challenge of Climate Change.

References

- [1] Environmental Protection Agency. *Ireland's Greenhouse Gas Emissions Projections*. Tech. rep. Environmental Protection Agency, 2024. URL: <https://www.epa.ie/publications/monitoring--assessment/climate-change/air-emissions/irelands-greenhouse-gas-emissions-projections-2023-2050.php>.
- [2] Environmental Protection Agency. *Climate Change in the Irish Mind Wave 2 Report 1*. Tech. rep. Environmental Protection Agency, 2024. URL: <https://www.epa.ie/publications/monitoring--assessment/climate-change/climate-change-in-the-irish-mind-wave-2-report-1.php>.
- [3] K. De Bruin, C. Kweku Kyei, and L. Henry. *Climate change impacts and associated economic costs in Ireland*. Tech. rep. Economic and Social Research Institute, 2024. URL: <https://www.esri.ie/publications/climate-change-impacts-and-associated-economic-costs-in-ireland>.
- [4] Government of Ireland. *Climate Action Plan 2024*. Tech. rep. Government of Ireland, 2024. URL: <https://www.gov.ie/en/publication/79659-climate-action-plan-2024/>.
- [5] Environmental Protection Agency. *Ireland's Provisional Greenhouse Gas Emissions*. Tech. rep. Environmental Protection Agency, 2024. URL: <https://www.epa.ie/publications/monitoring--assessment/climate-change/air-emissions/irelands-provisional-greenhouse-gas-emissions-1990-2023.php>.
- [6] Environmental Protection Agency. *Climate Change in the Irish Mind Wave 1 Report 1*. Tech. rep. Environmental Protection Agency, 2021. URL: <https://www.epa.ie/publications/monitoring--assessment/climate-change/climate-change-in-the-irish-mind.php>.
- [7] Commission to the European Parliament and the Council. *on the functioning of the European carbon market in 2022 pursuant to Articles 10(5) and 21(2) of Directive 2003/87/EC*. Tech. rep. European Commission, 2023. URL: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52023DC0654>.
- [8] Transport for Ireland. *People now using TFI Local Link services 100,000 times a week in rural Ireland*. WebPage. 2024. URL: <https://www.transportforireland.ie/news/people-now-using-tfi-local-link-services-100000-times-a-week-in-rural-ireland/#:~:text=three%20years%20now.,The%20weekly%20national%20average%20was%2016%2C745%20in%202021%3B%2036%2C664%20in,Mobility%20Plan%2C%20from%20January%202022>.